Monotone Function Intervals

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Motivation: Distributions of Posterior Quantiles

Limits of Gerrymandering

What kinds of legislatures can be made if any electoral map can be drawn?

A continuum of citizens, single-peaked preferences over political positions

- Ideal positions $x \in \mathbb{R}$ follow common prior CDF F.

Electoral map: signal μ, segments the population

- Each *district* $G \in \text{supp}(\mu)$, distribution of citizens' ideal positions within that specific district

Motivation: Distributions of Posterior Quantiles

Limits of Gerrymandering

Median Voter Theorem $\rightarrow$ The ideal position of the elected *representative* for district G is its median.

- All representatives' ideal positions follow the distribution $H^{1/2}(\cdot \mid \mu)$ of posterior medians.

Q: *How can we characterize all such $H^{1/2}$?*

Motivation: Distributions of Posterior Quantiles

Quantile-Based Persuasion

A state $x \in \mathbb{R}$ drawn from a common prior distribution F.

A sender commits to a signal to reveal information to a receiver.

- Receiver then chooses an action based on the realization of the signal.

KG (2011) proposes Concavification as the solution.

- $|\text{supp}(F)| > 2$ makes this method computationally challenging.

Motivation: Distributions of Posterior Quantiles

Quantile-Based Persuasion

Tractable reduction: Posterior-Mean approach (and majorization)

- The only payoff-relevant statistic of a posterior is its mean.
- Distributions of posterior means are mean-preserving contractions (MPCs) of the prior.
  - Gentzkow and Kamenica (2016); Kleiner, Moldovanu and Philipp Strack (2021)
  - Roesler and Szentes (2017); Ravid, Roesler and Szentes (2022)

A natural extension: from mean-based to quantile-based

Q: *How can we characterize distributions of posterior quantiles?*

Signals and Posterior Quantiles

One-dimensional state $x \in \mathbb{R}$ drawn from a prior CDF $F \in \mathcal{F}_0$.

Signal $\mu \in \Delta(\mathcal{F}_0)$, distribution over posteriors G with martingale property:

$$\int_{\mathcal{F}_0} G(x)\,\mu(dG) = F(x), \quad \forall x \in \mathbb{R}.$$

Target quantile $\tau \in (0,1)$

- Quantile function $G^{-1}(\tau) := \inf\{x \in \mathbb{R} : G(x) \geq \tau\}$.
- *τ-quantile selection rule* $r \in \mathcal{R}_\tau$, $\mathcal{R}_\tau := \Delta([G^{-1}(\tau), G^{-1}(\tau^+)])$.

$\mathcal{H}_\tau$: the set of distributions of posterior τ-quantiles induced by some signal μ and selection rule r.

Extreme Cases: Truncations

Define

$$F_L^\tau := \min\left\{\frac{1}{\tau}F(x), 1\right\}, \qquad F_R^\tau := \max\left\{\frac{F(x)-\tau}{1-\tau}, 0\right\}$$

to be the *left truncation* and the *right truncation* of the prior.

- F_L^τ (F_R^τ) is the conditional distribution of F given that x is smaller (larger) than a τ-quantile.

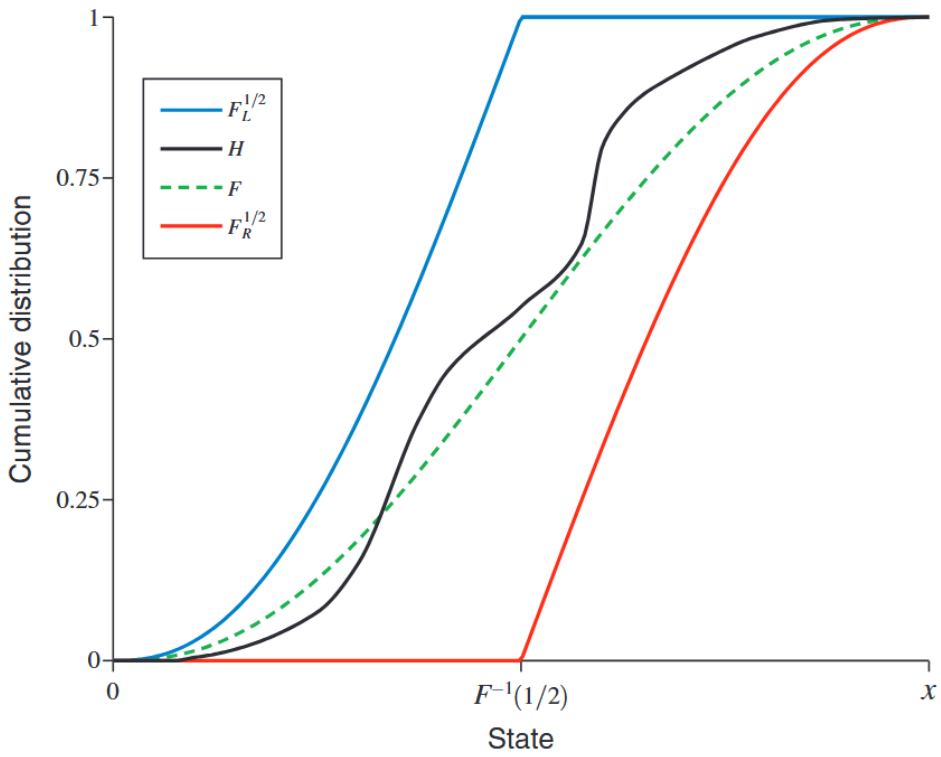
By "continuity" of the mapping from (μ, r) to the CDF of posterior τ-quantiles, we conjecture that:

Distributions of Posterior Quantiles

Theorem 2 (Distributions of Posterior Quantiles)

For any $\tau \in (0,1)$,

$$\mathcal{H}_\tau = \{H \in \mathcal{F}_0 : F_R^\tau(x) \leq H(x) \leq F_L^\tau(x),\ \forall x \in \mathbb{R}\}.$$



Proof of Theorem 2

Proof. *Step 1* $\mathcal{H}_\tau$ is a subset of the set on the RHS.

Let $H \in \mathcal{H}_\tau$ be induced by μ. Fix a state x.

- $H(x) = \mu(\mathcal{G}_x)$, where $\mathcal{G}_x \subseteq \text{supp}(\mu)$ is the set of posteriors whose τ-quantiles are weakly less than x.

For all $G \in \mathcal{G}_x$, let q_G be its τ-quantile: $q_G \leq x$, $\tau \leq G(q_G) \leq G(x)$.

- By the martingale property,

$$F(x) = \int_{\mathcal{F}_0} G(x)\,\mu(dG) \geq \int_{\mathcal{G}_x} G(x)\,d\mu \geq \tau\,\mu(\mathcal{G}_x) = \tau H(x).$$

  Thus we have that $H(x) \leq F_L^\tau(x)$.

Proof of Theorem 2 (Cont.)

Proof. *Step 2* The set on the RHS is a subset of $\mathcal{H}_\tau$.

- Need to construct a signal and a selection rule for each element in $\mathcal{H}_\tau$.
- Consider all functions in $\mathcal{H}_\tau$? HARD!

Luckily, extreme points still work.

Q: *How can we characterize the extreme points of this set, or more generally, the set of functions lying pointwise between two monotone functions?*

Monotone Function Interval

Let $\mathcal{F}$ be the set of nondecreasing and right-continuous functions on $\mathbb{R}$.[1]

For any $\underline{F}, \overline{F} \in \mathcal{F}$ such that $\underline{F}(x) \leq \overline{F}(x)$, $\forall x \in \mathbb{R}$ (henceforth $\underline{F} \leq \overline{F}$), let

$$\mathcal{I}(\underline{F}, \overline{F}) := \{H \in \mathcal{F} : \underline{F} \leq H \leq \overline{F}\},$$

which we refer to as **monotone function interval** bounded by $\underline{F}$ and $\overline{F}$.

[1] This space is endowed with the topology defined by weak (pointwise) convergence and the corresponding Borel σ-algebra.

Extreme Points of Monotone Function Intervals

Theorem 1 (Extreme Points of Monotone Function Intervals)

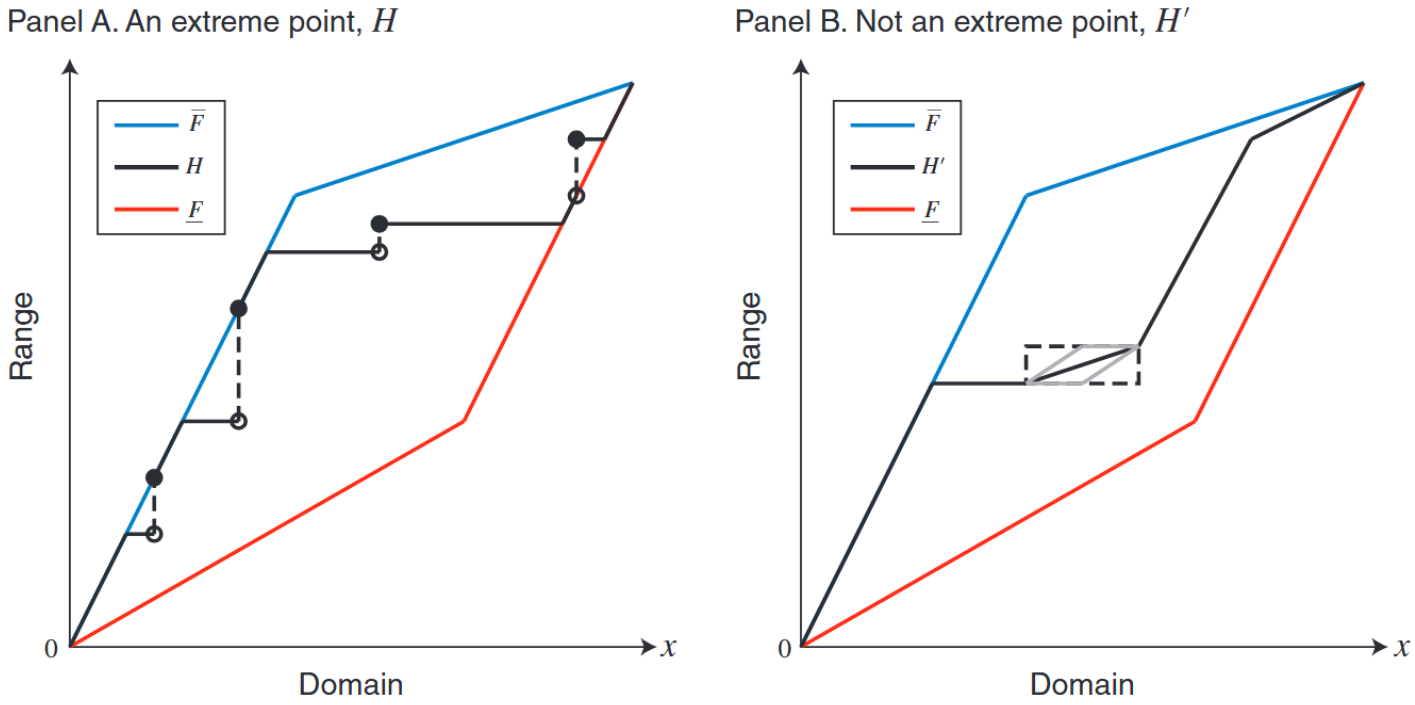
For any $\underline{F}, \overline{F} \in \mathcal{F}$ such that $\underline{F} \leq \overline{F}$, H is an extreme point of $\mathcal{I}(\underline{F}, \overline{F})$ if and only if there exists a countable collection of intervals $\{[\underline{x}_n, \overline{x}_n)\}_{n=1}^{\infty}$ such that

1. *$H(x) \in \{\underline{F}(x), \overline{F}(x)\}$ for all $x \notin \bigcup_{n=1}^{\infty} [\underline{x}_n, \overline{x}_n)$;*
2. *For all $n \in \mathbb{N}$, H is constant on $[\underline{x}_n, \overline{x}_n)$, and either $H(\overline{x}_n^-) = \underline{F}(\overline{x}_n^-)$ or $H(\underline{x}_n) = \overline{F}(\underline{x}_n)$.*

Extreme Points of Monotone Function Intervals

Proof. Sufficiency is straightforward.

Necessity: There exists a rectangle that lies between the graphs of $\underline{F}$ and $\overline{F}$, so that when restricted to this rectangle, the graph of H' is not a step function with only one jump.



Remarks on Theorem 1

Remarks

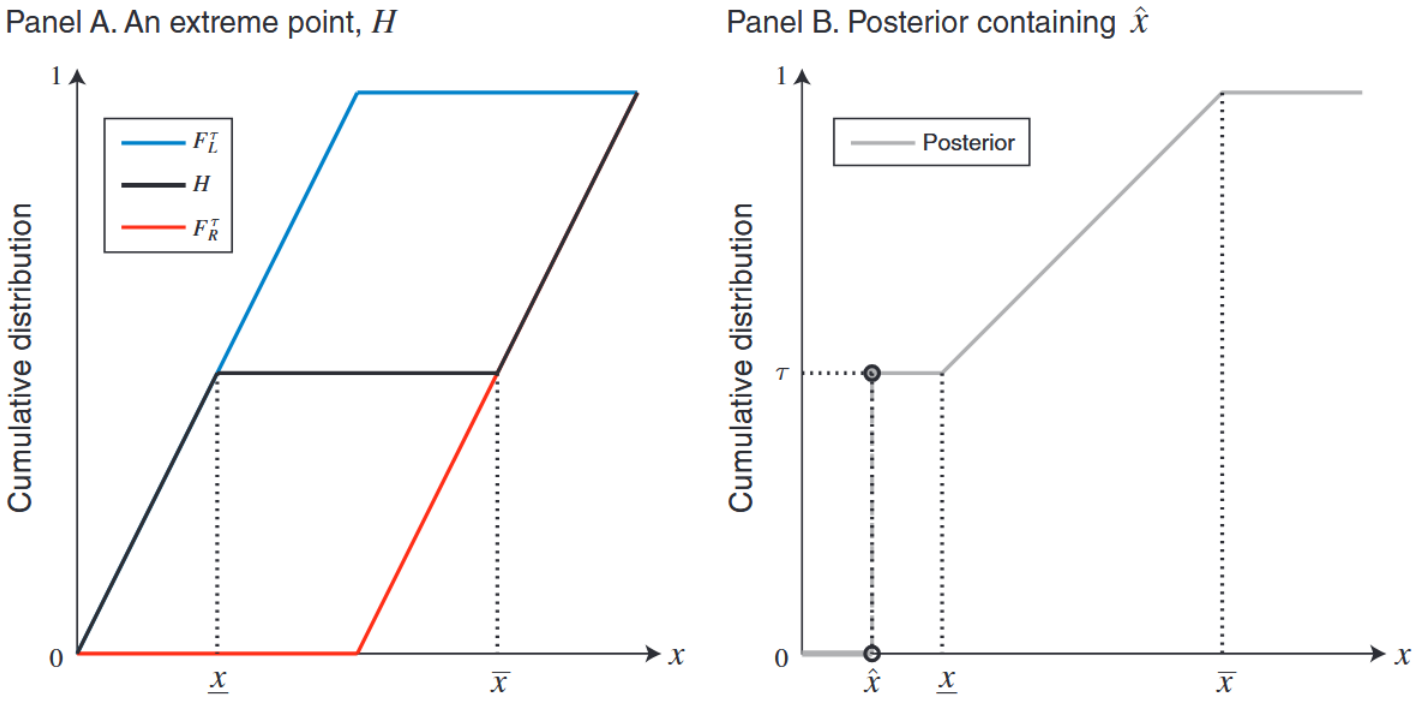
1. Only real domain?
2. Only nondecreasing?
3. Only monotone?
4. Can $\underline{F}$ and $\overline{F}$ take infinity values?

Theorem 2 Revisited

Theorem 2 (Distributions of Posterior Quantiles)

For any $\tau \in (0,1)$,

$$\mathcal{H}_\tau = \{H \in \mathcal{F}_0 : F_R^\tau(x) \leq H(x) \leq F_L^\tau(x),\ \forall x \in \mathbb{R}\}.$$



Theorem 2 Revisited

Construction of μ for extreme points:

1. Isolating atoms
   - Mass η at $\hat{x}$. Construct $\hat{F}$ such that $\hat{F}^{-1}(\tau) = \hat{x}$ to capture the mass nearby.
   - Decomposition $F = \eta\hat{F} + (1-\eta)\widetilde{F}$.
2. Avoiding flat intervals
   - H constant on some $(\underline{x}, \overline{x})$. No posterior τ-quantiles inside.
   - Split $\widetilde{F}$ into $\widetilde{F}_1$ and $\widetilde{F}_2$: $\widetilde{F}_1^{-1}(\tau) \leq \underline{x}$, $\widetilde{F}_2^{-1}(\tau) \geq \overline{x}$.

Distributions of Unique Posterior Quantiles

The result remains (essentially) the same when restricted to signals inducing a unique posterior τ-quantile.

Theorem 3 (Distributions of Unique Posterior Quantiles)

For any $\tau \in (0,1)$ and for any $F \in \mathcal{F}_0$ that has a full support on an interval,

$$\bigcup_{\varepsilon>0} \mathcal{I}(F_R^{\tau,\varepsilon}, F_L^{\tau,\varepsilon}) \subseteq \widetilde{\mathcal{H}}_\tau \subseteq \mathcal{I}(F_R^{\tau}, F_L^{\tau}),$$

where $\left\{\mathcal{I}(F_R^{\tau,\varepsilon}, F_L^{\tau,\varepsilon})\right\}_{\varepsilon>0}$ is a family of perturbations of $\mathcal{I}(F_R^{\tau}, F_L^{\tau})$:

$$F_L^{\tau,\varepsilon} := \frac{1}{\tau+\varepsilon} F(x)\mathbf{1}\{x < F^{-1}(\tau)\} + \mathbf{1}\{x \geq F^{-1}(\tau)\},$$

$$F_R^{\tau,\varepsilon} := \frac{F(x) - (\tau - \varepsilon)}{1 - (\tau - \varepsilon)} \mathbf{1}\{x \geq F^{-1}(\tau)\}.$$

Law of Iterated Quantiles

Corollary 1 (Law of Iterated Quantiles)

Consider any $\tau, q \in (0,1)$.

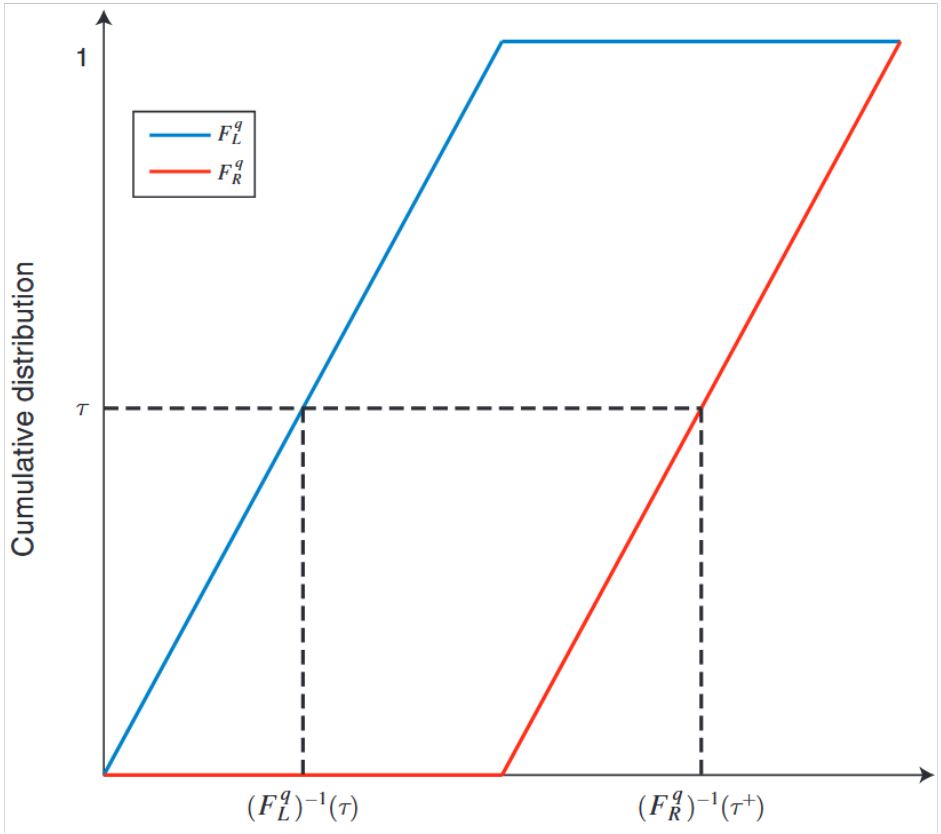
1. *For any closed interval* $Q \subseteq \mathbb{R}$, $Q = [H^{-1}(\tau), H^{-1}(\tau^+)]$ *for some* $H \in \mathcal{H}_q$ *if and only if*

$$Q \subseteq [(F_R^q)^{-1}(\tau), (F_L^q)^{-1}(\tau^+)].$$

2. *Suppose that the prior* F *is continuous and has full support on an interval. Then for any* $\hat{x} \in \mathbb{R}$, $\hat{x} \in [H^{-1}(\tau), H^{-1}(\tau^+)]$ *for some* $H \in \widetilde{\mathcal{H}}_q$ *if and only if*

$$\hat{x} \in [(F_R^q)^{-1}(\tau), (F_L^q)^{-1}(\tau)].$$

Law of Iterated Quantiles



The τ-quantiles of posterior q-quantiles coincide with the interval $[(F_R^q)^{-1}(\tau), (F_L^q)^{-1}(\tau^+)]$.

- An analog of the law of iterated expectations

Gerrymandering Revisited

Proposition 1 (Limits of Gerrymandering)

For any $H \in \mathcal{F}_0$, the following are equivalent:

1. $H \in \mathcal{I}\left(F_R^{1/2}, F_L^{1/2}\right)$.
2. *H is a distribution of the representatives' ideal positions under some map $\mu \in \mathcal{M}$ and some selection rule $r \in \mathcal{R}_{1/2}$.*

If, furthermore, $F \in \mathcal{F}_0$ has full support on an interval, then for any fixed selection rule $\hat{r} \in \mathcal{R}_{1/2}$, every

$$H \in \bigcup_{\varepsilon > 0} \mathcal{I}\left(F_R^{1/2,\,\varepsilon}, F_L^{1/2,\,\varepsilon}\right)$$

is a distribution of the representatives' ideal positions under some map $\mu \in \mathcal{M}$ and selection rule $\hat{r}$.

Gerrymandering Revisited

Legislative outcomes: Assume legislation is a median of representatives.

Corollary 2 (Limits of Legislative Outcomes)

Suppose that the median voter property holds both at the district and legislative level. Then an outcome $x \in \mathbb{R}$ can be enacted as legislation under some map if and only if

$$x \in \left[F^{-1}(1/4), F^{-1}(3/4^{+})\right].$$

Quantile-Based Persuasion Revisited

Proposition 2 (Quantile-Based Persuasion)

Suppose that the prior F has full support on some interval, and suppose that there exists $\tau \in (0,1)$, a selection rule $r \in \mathcal{R}_\tau$, and a measurable function $v_S : \mathbb{R} \to \mathbb{R}$ such that

$$\hat{v}(G) = \int_{\mathbb{R}} v_S(x)\, r(dx \mid G), \qquad \forall G \in \mathcal{F}_0.$$

Then

$$\operatorname{cav}(\hat{v})[F] \;=\; \sup_{H \in \operatorname{ext}\left(\mathcal{I}(F_R^\tau, F_L^\tau)\right)} \int_{\mathbb{R}} v_S(x)\, H(dx).$$

Given some assumptions on v_S, one can characterize the maximizers.

Apparent Misconfidence

Benoît and Dubra (2011)

Apparent misconfidence
The distribution of individuals' self-predictions deviates from the true population distribution.

- *When do empirical observations of apparent misconfidence imply irrationality?*
- *Can such data be rationalized by Bayesian agents receiving noisy signals and reporting the τ-quantile of their posterior beliefs?*

Apparent Misconfidence

A unit mass of individuals, each attached to a type $x \in [0, 1]$.

- Their skill levels follow a CDF F.
- Given a finite partition $0 = z_0 < z_1 < \cdots < z_K = 1$, a *prediction dataset* is a vector $(\theta_k)_{k=1}^K$ with $\sum_{k=1}^K \theta_k = 1$.
  - θ_k: share of individuals who predict their own types in $[z_{k-1}, z_k)$.

Apparent Misconfidence

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  - θ_k: share of individuals who predict their own types in $[z_{k-1}, z_k)$.
- $(\theta_k)_{k=1}^K$ is τ-*quantile rationalizable* (i.e., consistent with Bayesian belief and reporting τ-quantiles) if there is a signal such that
  1. the induced posterior has a uniqe τ-quantile with prob. 1;
  2. for all k, the prob. of the τ-quantile being in $[z_{k-1}, z_k)$ is θ_k.

Rationalizable Apparent Misconfidence

Corollary 5 (Rationalizable Apparent Misconfidence)

For any $\tau \in (0,1)$, for any $F \in \mathcal{F}_0$ with full support on $[0,1]$, and for any partition $0 = z_0 < z_1 < \cdots < z_K = 1$ of $[0,1]$, a prediction dataset $(\theta_k)_{k=1}^K$ is τ-quantile rationalizable if and only if for all $k \in \{1, \ldots, K\}$,

$$\sum_{i=1}^{k} \theta_i < \frac{1}{\tau} F(z_k), \qquad \sum_{i=k}^{K} \theta_i < \frac{1 - F(z_{k-1}^-)}{1 - \tau}.$$

Security Design with Moral Hazard

A risk-neutral entrepreneur (he) issues a security to an investor (she) to fund a project.

- Investment $I > 0$ needed for the project
- If funded, he exerts effort $e \in [0, \overline{e}]$.
  - Project's profit follows $\Phi(\cdot \mid e) \in \mathcal{F}_0$.
  - Effort cost $C(e) \geq 0$

Security specifies the return to the investor for each realized profit $x \geq 0$.

- *Limited liability*: Security is a function $H : \mathbb{R}_+ \to \mathbb{R}$, $0 \leq H \leq x$.
- A security is also required to be monotone in the profit.

Security Design with Moral Hazard

The entrepreneur, given H, chooses e to solve

$$\sup_{e\in[0,\overline{e}]} \int_0^\infty (x - H(x))\, \Phi(dx \mid e) - C(e).$$

The entrepreneur's goal is to design a security to acquire funding from the investor while maximizing her expected payoff:

$$\sup_{H\in \mathrm{ext}(\mathcal{I}(0,\mathrm{id}))\,,\ e\in[0,\overline{e}]} \left[\int_0^1 [x - H(x)]\phi(x \mid e)\, dx - C(e)\right]$$

$$\text{s.t.} \qquad \int_0^1 [x - H(x)]\phi_e'(x \mid e)\, dx = C'(e),$$

$$\int_0^1 H(x)\phi_e'(x \mid e)\, dx \geq (1+r)I,$$

where $r > 0$ is the return rate of a risk-free asset.

Security Design with Moral Hazard

Innes (1990): With $\{\phi(\cdot \mid e)\}_{e \in [0,\overline{e}]}$ satisfying the *monotone likelihood ratio* property (Milgrom 1981), every optimal security must be

$$H^d(x) := \min\{x, d\}$$

for some face value $d > 0$.

Extreme Points of $\mathcal{I}(0, \mathrm{id})$

Theorem 4 (Extreme Points of $\mathcal{I}(0, \mathrm{id})$ Bounded by Linear Functionals)

For any $J \in \mathbb{N}$, for any collection of bounded linear functionals $\{\Gamma_j\}_{j=1}^J$ on $L^1([0,1])$ and for any collection $\{\gamma_j\}_{j=1}^J \subseteq \mathbb{R}$, let $\mathcal{I}^c$ be a convex subset of $\mathcal{I}(0, id)$ defined as

$$\mathcal{I}^c := \{H \in \mathcal{I}(0, id) : \Gamma_j(H) \geq \gamma_j,\ \forall j \in \{1, \ldots, J\}\}.$$

Suppose that $H \in \mathcal{I}^c$ is an extreme point of $\mathcal{I}^c$. Then there exists countably many intervals $\{[\underline{x}_n, \overline{x}_n)\}_{n=1}^\infty$ such that:

1. *$H(x) = x$ for all $x \notin \bigcup_{n=1}^\infty [\underline{x}_n, \overline{x}_n)$;*
2. *For all $n, m \in \mathbb{N}$, with $n \neq m$, H is constant on $[\underline{x}_n, \overline{x}_n)$ and $H(\underline{x}_n) \neq H(\underline{x}_m)$;*
3. *For all but at most J many $n \in \mathbb{N}$, $H(\underline{x}_n) = \underline{x}_n$.*

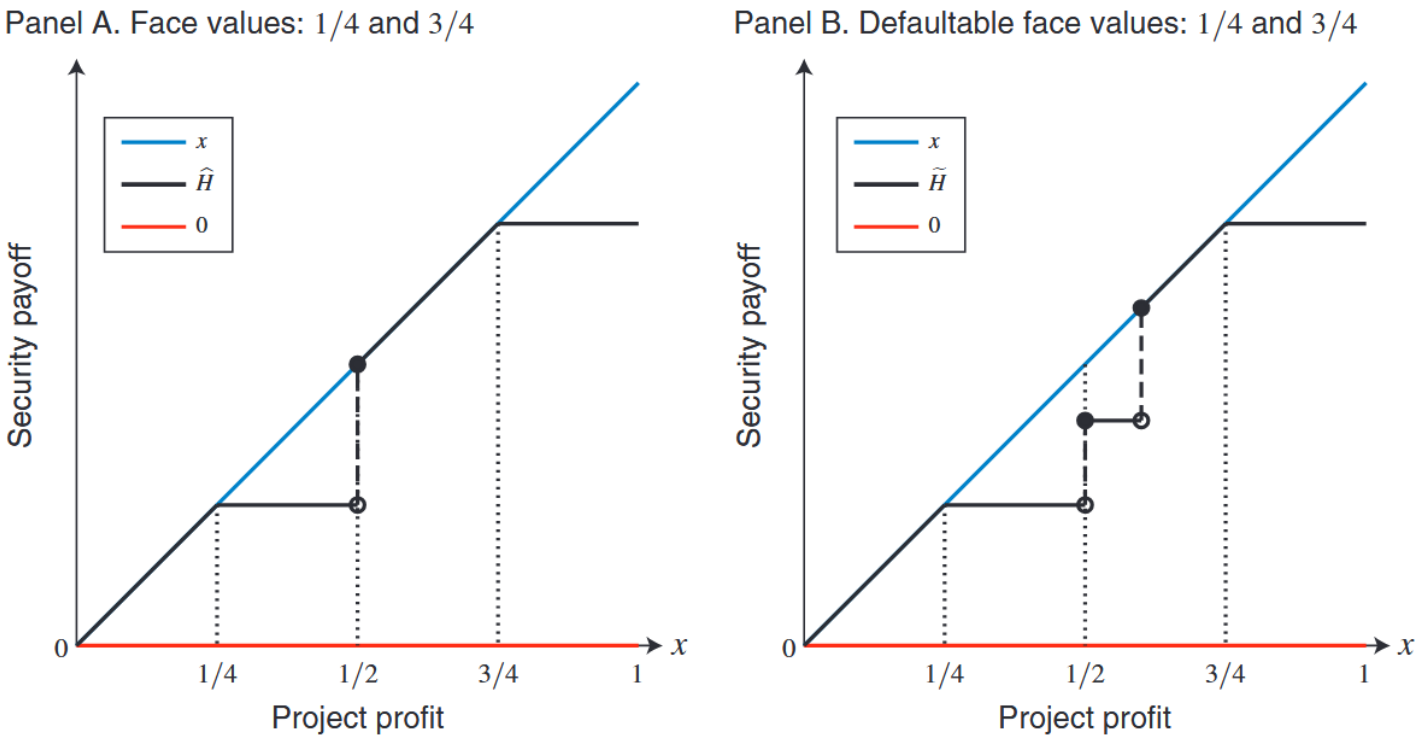
Contingent Debt Contracts

A security $H \in \mathcal{I}(0, \mathrm{id})$ is a *contingent debt contract* with N possible face values; i.e., if there exists $\{[\underline{x}_n, \overline{x}_n)\}_{n \geq 1}$ such that:

1. H is constant on $[\underline{x}_n, \overline{x}_n)$;
2. $H(x) = x$ for all $x \notin \bigcup_{n=1}^{\infty} [\underline{x}_n, \overline{x}_n)$;
3. face values $H(\underline{x}_n) \neq H(\underline{x}_m)$ for all $n \neq m$.

A face value $d_n = H(\underline{x}_n)$ is *nondefaultable* if $d_n < x$ for all $x \in [\underline{x}_n, \overline{x}_n)$.

Contingent Debt Contracts



Proposition 3

There is a contingent debt contract with at most two nondefaultable face values that solves the entrepreneur's problem.

Security Design with Adverse Selection

A risk-neutral security issuer (she), a unit mass of risk-neutral investors

- An asset generates a random cash flow $x \geq 0$ in period 1.
  - Cash flow follows $\Phi_0 \in \mathcal{F}_0$ supported on $[0, 1]$.
- The issuer sells in period 0 a limited-liability security H
  - She first observes a signal $s \in S$ for the cash flow.

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DeMarzo and Duffie (1999): In the unique equilibrium, the expected value of H is

$$\Pi(H) := (1-\delta)\left(\inf_{s\in S}\int_0^1 H(x)\,\Phi(dx \mid s)\right)^{1/(1-\delta)} \int_S \left(\int_0^1 H(x)\,\Phi(dx \mid s)\right)^{-\delta/(1-\delta)} \Psi(ds).$$

The security design problem:

$$\sup_{H\in\mathcal{I}(0,\mathrm{id})} \Pi(H).$$

Security Design with Adverse Selection

DeMarzo and Duffie (1999) restrict attention to signal structure $\{\Phi(\cdot \mid s)\}_{s \in S}$ with a *uniform worst case* s_0.

- We can slightly relax this to an overall worst case s_0.

Proposition 5

There is a contingent debt contract with at most one nondefaultable face value that solves the issuer's problem. Furthermore, if $\Phi(\cdot \mid s)$ *has full support on* $[0, 1]$ *for all* $s \in S$*, this solution is unique.*

Takeaways

- Monotone function intervals have simple extreme points.
  - An extreme point either coincides with a bound, or is constant on an interval with at least one endpoint on the a bound.
  - Provide an extreme point approach to handle (i) FOSD, (ii) limited liability in contracts, (iii) outward/inward shifts of supply/demand...
- All posterior quantile distributions form the interval between the "all-left" and "all-right" truncations.
- Techniques for information/mechanism design
  (a) Basic properties (martingales)
  (b) Reduction (posterior moments/quantiles, assumptions on distributions)
  (c) Optimization (concavification, extreme points, duality)